SG No. 195 T^1 $P23$ [cubic]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `1a`, site symmetry: `23.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `3c`, site symmetry: `222..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12]`

Table 4: Wyckoff site: `3d`, site symmetry: `222..`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[5,6,7,8]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[9,10,11,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,9]`
2	$[-x, -x, x]$	`[2,7,12]`
3	$[-x, x, -x]$	`[3,8,10]`
4	$[x, -x, -x]$	`[4,6,11]`

Table 6: Wyckoff site: `6f`, site symmetry: `2..`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4]`
2	$[-x, 0, 0]$	`[2,3]`
3	$[0, x, 0]$	`[5,8]`
4	$[0, -x, 0]$	`[6,7]`
5	$[0, 0, x]$	`[9,12]`

6	$[0,\, 0,\, -x]$	`[10,11]`

Table 7: Wyckoff site: `6g`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[1,4]`
2	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2},\, x,\, 0\right]$	`[5,8]`
4	$\left[\frac{1}{2},\, -x,\, 0\right]$	`[6,7]`
5	$\left[0,\, \frac{1}{2},\, x\right]$	`[9,12]`
6	$\left[0,\, \frac{1}{2},\, -x\right]$	`[10,11]`

Table 8: Wyckoff site: `6h`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	`[1,4]`
2	$\left[-x,\, \frac{1}{2},\, 0\right]$	`[2,3]`
3	$\left[0,\, x,\, \frac{1}{2}\right]$	`[5,8]`
4	$\left[0,\, -x,\, \frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{2},\, 0,\, x\right]$	`[9,12]`
6	$\left[\frac{1}{2},\, 0,\, -x\right]$	`[10,11]`

Table 9: Wyckoff site: `6i`, site symmetry: `2..`

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[1,4]`
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[5,8]`
4	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{2},\, \frac{1}{2},\, x\right]$	`[9,12]`
6	$\left[\frac{1}{2},\, \frac{1}{2},\, -x\right]$	`[10,11]`

Table 10: Wyckoff site: `12j`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-x,\, -y,\, z]$	`[2]`
3	$[-x,\, y,\, -z]$	`[3]`
4	$[x,\, -y,\, -z]$	`[4]`
5	$[z,\, x,\, y]$	`[5]`
6	$[z,\, -x,\, -y]$	`[6]`
7	$[-z,\, -x,\, y]$	`[7]`
8	$[-z,\, x,\, -y]$	`[8]`
9	$[y,\, z,\, x]$	`[9]`

continued ...

Table 10

No.	position	mapping
10	$[-y, z, -x]$	`[10]`
11	$[y, -z, -x]$	`[11]`
12	$[-y, -z, x]$	`[12]`